

pyDRT-Peak-Analysis

Installation, Launch, and User Guide

Step-by-step instructions for installation, EIS analysis, peak deconvolution, export, and citation

What this guide does

This document explains, from the beginning, how to install Python, download the project from GitHub, install the required Python packages, start the graphical user interface (GUI), and fix the most common installation problems.

Project repository: <https://github.com/MFakouri/pyDRT-Peak-Analysis->

Repository version identified in setup.py: 0.4.0

Contents

1. Before you start	3
2. Recommended Windows installation (easiest method)	4
3. Starting the program after installation	6
4. Manual installation using Command Prompt	7
5. macOS and Linux installation	8
6. How to update to a newer version	8
7. Common problems and fixes	9
8. Quick verification checklist	10
9. Technical notes and source files checked	11
10. Using pyDRT-Peak-Analysis: step-by-step workflow	12
11. Contact and citation	16

Fastest path for most Windows users

Install Python → download the GitHub ZIP → extract the ZIP → double-click install_windows.bat once → double-click run_windows.bat whenever you want to use the program.

1. Before you start

You do not need to know Python, Git, GitHub, or the Command Prompt to use the recommended Windows method in this guide.

You need:

- **A Windows computer.** Windows 10 or Windows 11 is the simplest environment for the included .bat helper files.
- **An internet connection.** It is required during the first installation so Python and the required packages can be downloaded.
- **Python 3.9 or newer.** The project setup file specifies Python 3.9+.
- **Enough free disk space.** Python plus scientific packages require additional storage beyond the project ZIP itself.
- **No GitHub account is required.** You can download the public repository as a ZIP file without signing in.

Important

Do not try to run the project while it is still inside the downloaded ZIP file. Extract the ZIP first. The installation and launch files expect the project files to remain together in the same folder.

What the project already includes

- **install_windows.bat** — installs or updates pip and then installs everything listed in requirements.txt.
- **run_windows.bat** — starts the application by running launch.py.
- **launch.py** — launches the pyDRTtools graphical user interface.
- **requirements.txt** — lists NumPy, SciPy, Pandas, Matplotlib, scikit-learn, PyQt5, and Click as required packages.

CVXOPT is optional in this fork

The current qp_solver.py uses CVXOPT when it is available, but it also includes a SciPy optimization fallback. CVXOPT is therefore not listed in requirements.txt and is not required for the basic installation described here.

2. Recommended Windows installation (easiest method)

Recommended approach

Use the two included Windows batch files. This avoids typing package-installation commands manually.

Step 1. Install Python

Open the official Python for Windows download page: python.org/downloads/windows

1. Download a normal 64-bit Windows Python installer or use the official Python install manager.
2. Run the installer you downloaded.
3. If the installer shows an option such as “Add python.exe to PATH”, enable it. This makes Python easier to use from Command Prompt.
4. Complete the Python installation.

Python version

The repository requires Python 3.9 or newer. If you later encounter package-compatibility problems with a very new Python release, installing Python 3.11 or 3.12 is a reasonable compatibility fallback.

Step 2. Check that Python is installed correctly

1. Press the Windows key on your keyboard.
2. Type “cmd”.
3. Open Command Prompt.
4. Type the command below and press Enter.

```
py --version
```

You should see a result similar to: **Python 3.x.x**. The exact last numbers can be different.

If Windows says “py is not recognized”

Do not continue to the package-installation step yet. Reinstall Python and make sure the Python launcher/PATH options are enabled, or use “python --version” instead. See Section 7 for detailed fixes.

Step 3. Download the project from GitHub

Open the project page: <https://github.com/MFakouri/pyDRT-Peak-Analysis->

1. On the GitHub page, click the green “Code” button near the top of the file list.
2. Choose “Download ZIP”.
3. Wait for the ZIP file to finish downloading. It will normally appear in your Downloads folder.

Step 4. Extract the ZIP file

1. Open your Downloads folder in File Explorer.
2. Find the downloaded ZIP file for pyDRT-Peak-Analysis.
3. Right-click the ZIP file and choose “Extract All...”.
4. Choose a simple location such as Documents or Desktop.
5. Click “Extract”.

Recommended folder location

Use a normal user folder such as Documents or Desktop. Avoid protected system folders such as C:\Program Files, because Windows may require administrator permissions there.

Step 5. Open the extracted project folder

Open the extracted folder. At the top level, you should see files with names similar to:

- install_windows.bat

- run_windows.bat
- launch.py
- requirements.txt
- setup.py
- README.md
- a folder named pyDRTtools

If you only see another folder with the same project name

Open that folder too. You are in the correct location when install_windows.bat, run_windows.bat, launch.py, and requirements.txt are visible together.

Step 6. Run the one-time installer

1. Double-click install_windows.bat.
2. A black Command Prompt window will open.
3. The script first upgrades pip and then installs the packages listed in requirements.txt.
4. During installation, text will scroll in the black window. This is normal.
5. When installation is finished, the window shows a “Press any key to continue...” message because the script contains a pause command.
6. Press any key to close the window.

Do not close the installation window early

The first installation can download several scientific Python packages. Wait until the script reaches the final pause or clearly reports an error.

What install_windows.bat actually runs

```
py -m pip install --upgrade pip
py -m pip install -r requirements.txt
```

Step 7. Start pyDRT-Peak-Analysis

1. After installation completes, stay in the same extracted project folder.
2. Double-click run_windows.bat.
3. A Command Prompt window may appear briefly.
4. The Python graphical user interface should open.

What run_windows.bat actually runs

```
py launch.py
```

Installation is only needed once

Normally, you run install_windows.bat only the first time (or after updating the project/requirements). For everyday use, open the project folder and double-click run_windows.bat.

3. Starting the program after installation

After the first successful installation, starting the application is simple:

1. Open File Explorer.
2. Go to the extracted pyDRT-Peak-Analysis folder.
3. Double-click run_windows.bat.
4. Wait for the GUI window to appear.

What you should expect

- The launcher imports the GUI from the pyDRTtools package and calls its launch function.
- A black terminal window can remain open while the GUI is running. Do not close it unless you also want to stop the program.
- If an error occurs, the terminal text is useful. Take a screenshot or copy the last several lines before closing it.

A useful habit

Keep the entire project folder intact. Do not move launch.py, run_windows.bat, or the pyDRTtools folder separately, because the launcher depends on the project's folder structure.

Optional: create a desktop shortcut

1. Right-click run_windows.bat.
2. Choose "Show more options" if Windows 11 displays the compact menu.
3. Choose "Send to" → "Desktop (create shortcut)" if that option is available.
4. Use the new shortcut for future launches. Keep the original project folder in place.

4. Manual installation using Command Prompt

Use this method if double-clicking install_windows.bat does not work, or if you want to see exactly which command fails.

Step 1. Open Command Prompt directly inside the project folder

1. Open the extracted project folder in File Explorer.
2. Click once in the File Explorer address bar at the top of the window.
3. Type cmd and press Enter.
4. A Command Prompt window should open already pointed at the project folder.

Step 2. Confirm Python works

```
py --version
```

If that command fails, try:

```
python --version
```

Step 3. Upgrade pip

```
py -m pip install --upgrade pip
```

If your computer uses the command “python” instead of “py”, use:

```
python -m pip install --upgrade pip
```

Step 4. Install the required packages

```
py -m pip install -r requirements.txt
```

This installs the packages defined by the project, including:

- NumPy >= 1.24
- SciPy >= 1.10
- Pandas >= 1.5
- Matplotlib >= 3.7
- scikit-learn >= 1.2
- PyQt5 >= 5.15.9
- Click >= 8.1

Step 5. Launch the program

```
py launch.py
```

If you used “python” in the previous steps, use:

```
python launch.py
```

How to know where the problem is

If package installation fails, the important information is usually near the bottom of the terminal output. Look for lines containing “ERROR”, “Failed”, “No matching distribution”, or “ModuleNotFoundError”.

4.1 Optional: install in an isolated virtual environment

This is not required for the easiest setup, but it can prevent conflicts with other Python projects. Use it if you already have Python software on the computer or if a normal installation causes dependency conflicts.

```
py -m venv .venv
.venv\Scripts\activate
python -m pip install --upgrade pip
python -m pip install -r requirements.txt
python launch.py
```

When the virtual environment is active, you normally see (.venv) at the beginning of the Command Prompt line.

5. macOS and Linux installation

Windows helper files do not apply here

install_windows.bat and run_windows.bat are Windows batch files. On macOS or Linux, use a Terminal and run the equivalent Python commands manually.

1. Install Python 3.9 or newer using the normal method for your operating system.
2. Download the repository ZIP from GitHub and extract it, or clone the repository if you already use Git.
3. Open Terminal.
4. Change directory to the extracted project folder.
5. Install the requirements and launch the program using the commands below.

```
python3 -m pip install --upgrade pip
python3 -m pip install -r requirements.txt
python3 launch.py
```

PyQt5 note

The GUI depends on PyQt5. On some Linux distributions, additional system-level Qt libraries may be needed. If the application installs but the GUI cannot start, check the Linux distribution's Qt/PyQt runtime requirements.

6. How to update to a newer version

The simplest update method for a non-Git user is to download a fresh ZIP from GitHub.

1. Close pyDRT-Peak-Analysis if it is running.
2. Go to the GitHub repository page.
3. Click "Code" → "Download ZIP".
4. Extract the new ZIP into a new folder. Keeping the old folder temporarily gives you an easy backup.
5. Open the new extracted folder.
6. Double-click install_windows.bat again. This ensures any new or changed dependencies are installed.
7. Double-click run_windows.bat and confirm the GUI opens.
8. After confirming the new version works, you may remove the old project folder if you no longer need it.

Do not mix old and new program files

For a clean update, replace the whole project folder rather than copying only a few Python files into an older folder. Mixed versions can cause difficult-to-diagnose errors.

7. Common problems and fixes

Problem 1: 'py' is not recognized

- Python or the Windows Python launcher is not available from Command Prompt.

- First try: `python --version`.
- If that works, use “python” instead of “py” in the manual commands.
- If neither command works, reinstall Python from python.org and enable the PATH/launcher options offered by the installer.

Problem 2: The .bat file opens and immediately closes

- Open Command Prompt inside the project folder using the address-bar method in Section 4.
- Run the commands manually. This keeps the terminal open so you can read the actual error.

Problem 3: ModuleNotFoundError

- A required Python package is missing or the installation used a different Python installation.
- Run: `py -m pip install -r requirements.txt`
- Then launch with the same Python command: `py launch.py`

Problem 4: No module named 'PyQt5'

- Run the requirements installation again.
- If it still fails, copy the exact PyQt5 installation error from the terminal. The cause can depend on the Python version and operating system.

Problem 5: No module named 'pyDRTtools'

- Make sure Command Prompt is opened in the top-level project folder.
- In that folder you should be able to see both `launch.py` and the `pyDRTtools` folder.
- Do not move `launch.py` away from the rest of the project.

Problem 6: Windows SmartScreen or security warning

- Confirm that the files were downloaded from the intended GitHub repository before proceeding.
- Windows may warn about downloaded script files. If you trust and verified the source, use the Windows option that allows the file to run. Wording varies by Windows version and security policy.
- On managed work computers, your organization may block batch scripts. In that case use the manual Command Prompt method or contact your IT administrator.

Problem 7: Permission denied / access denied

- Move the extracted project folder to Documents or Desktop instead of a protected system directory.
- Avoid `C:\Program Files` and other folders that normally require administrator privileges.
- Then rerun the installation.

Problem 8: Package installation fails on a very new Python version

- The project requires Python 3.9+, but third-party package support can lag behind the newest Python releases.
- If the terminal specifically reports that a compatible package/wheel cannot be found, install Python 3.11 or 3.12 and repeat the installation using that interpreter.

Problem 9: The GUI does not appear but no clear error is visible

- Run `py launch.py` manually from Command Prompt instead of double-clicking `run_windows.bat`.
- Keep the terminal open and inspect the last lines for an exception or error message.
- If you ask for support, include the Python version and the complete final error block.

Information to include when asking for help

- Your operating system (for example, Windows 11 64-bit).
- The output of: `py --version` (or `python --version`).
- Whether `install_windows.bat` completed successfully.
- The final 10–20 lines of the terminal error message.
- Whether you started the program with `run_windows.bat` or with `py launch.py`.

8. Quick verification checklist

Use this list after a new installation. If every item is true, the basic setup is complete.

1. Python 3.9 or newer is installed.
2. The repository ZIP has been extracted.
3. `install_windows.bat`, `run_windows.bat`, `launch.py`, `requirements.txt`, and the `pyDRTtools` folder are visible together.
4. `install_windows.bat` finished without an ERROR at the end.
5. `run_windows.bat` starts the program.
6. The `pyDRTtools` GUI window opens.

Everyday use

Once the installation is working, the normal routine is simply: open the project folder → double-click `run_windows.bat` → use the GUI.

9. Technical notes and source files checked

This guide was prepared from the public repository state reviewed on 25 August 2026. The installation instructions were cross-checked against the following project files:

File	Why it matters
README.md	States the basic installation commands: <code>pip install -r requirements.txt</code> and <code>python launch.py</code> .
setup.py	Project version 0.4.0; specifies Python ≥ 3.9 and the package dependencies.
requirements.txt	Lists the required Python packages and minimum versions.
install_windows.bat	Uses the Windows py launcher to upgrade pip and install requirements.txt.
run_windows.bat	Runs py launch.py.
launch.py	Imports <code>launch_gui</code> from <code>pyDRTtools.GUI</code> and starts the GUI.
pyDRTtools/qp_solver.py	Uses CVXOPT when available and otherwise falls back to SciPy optimization.

Repository: <https://github.com/MFakouri/pyDRT-Peak-Analysis->

Official Python downloads for Windows: <https://www.python.org/downloads/windows/>

10. Using pyDRT-Peak-Analysis: step-by-step workflow

This section explains the normal analysis workflow after the program has been installed and the GUI opens successfully. It is written for users who are not familiar with Python or command-line tools.

Recommended first run: For your first analysis, keep the default DRT settings unless you have a specific reason to change them. Import one EIS file, confirm that the Nyquist plot looks correct, run the Simple Run analysis, check the regression fit, and only then perform peak deconvolution.

Quick workflow

Step	Action	What to do
1	Start the GUI	Open run_windows.bat or run python launch.py.
2	Import EIS data	Click Import and select one EIS file.
3	Set active area	Apply area correction only when the impedance is still in Ω rather than $\Omega \cdot \text{cm}^2$.
4	Verify the import	Check the EIS Data/Nyquist plot for a physically sensible curve.
5	Run DRT	Under Simple Run, click Run.
6	Check the fit	Compare the fitted curve with the experimental EIS data and adjust parameters if needed.
7	Inspect DRT peaks	Open the DRT tab and count the visible peaks.
8	Deconvolute peaks	Enter the Number of peaks and click Run under Peak deconvolution.
9	Export results	Export DRT/EIS CSV files and save the figure/peak-parameter CSV.

Step 1 - Start the program

1. Open the extracted pyDRT-Peak-Analysis folder.
2. On Windows, double-click run_windows.bat. If you prefer Command Prompt, open the project folder and run: `python launch.py`
3. Wait for the pyDRTtools graphical window to open. The first start can take a little longer while Python loads the scientific libraries.

If nothing opens: Use the troubleshooting section earlier in this guide. Running `py launch.py` from Command Prompt is the best way to see the actual error message.

Step 2 - Import an EIS file

1. In the Settings panel, click Import.
2. Select the EIS file you want to analyze. The current file chooser accepts .mat, .txt, .csv, .z, and .dat files.
3. The importer looks for frequency, real-impedance, and imaginary-impedance columns. Common labels such as Freq/Frequency, Zreal/Z', Zimag/Z'', -Z'', Re(Z)/Ohm, and Im(Z)/Ohm are recognized.

The original workflow also uses direct exports from common EIS instruments such as Solartron, Autolab, Gamry, Admiral, and Zahner. Always confirm the imported curve visually before continuing, because vendor export settings can change column names or sign conventions.

Step 3 - Set the active-area correction correctly

After you choose a file, the Active Area dialog appears.

- If the impedance values are already reported in $\Omega \cdot \text{cm}^2$, leave “Active area has NOT already been applied” unchecked and click OK.
- If the impedance values are in Ω , check “Active area has NOT already been applied”, enter the electrode active area in cm^2 , and click OK.

Important: Do not apply the active area twice. If the imported data are already area-normalized and you apply the area again, all impedance-derived values will be scaled incorrectly.

Step 4 - Verify that the file was imported correctly

1. Open the EIS Data tab.
2. Check the Nyquist plot. The curve should look consistent with the original EIS data from your instrument or source file.
3. If the plot is missing, strongly distorted, mirrored unexpectedly, or clearly different from the source data, stop and check the input columns and units before running DRT.

Step 5 - Review the main DRT settings

For a first analysis, the current program starts with practical default values. A beginner can normally leave these unchanged for the first run:

Setting	Typical starting value
Method of Discretization	Gaussian
Data Used	Combined Re-Im
Inductance	Fitting with Inductance
Regularization derivative	2nd order
Parameter Selection Method	custom
Custom Regularization parameter	0.001
RBF Shape Control	FWHM Coefficient
FWHM Control	0.5
Number of Samples	2000 (used for Bayesian Run)

Note: These are starting values, not universal optimum values. The regularization parameter and FWHM Control can materially affect the DRT and should be checked against the EIS regression fit.

Step 6 - Run the DRT analysis and check the fit

1. Under Simple Run, click Run.
2. Wait for the calculation to finish, then return to the EIS Data tab.
3. Compare the fitted/regressed curve with the experimental EIS data. The fitted curve should follow the measured curve closely over the frequency range.
4. If the fit is not satisfactory, adjust the Regularization parameter and/or FWHM Control and run Simple Run again.

If a good fit cannot be achieved, especially at the very high- or very low-frequency ends, the EIS data may not be sufficiently reliable for the intended interpretation. In that situation, review the test conditions, wiring, frequency range, stability, and measurement setup and consider repeating the experiment after appropriate modifications.

Step 7 - Inspect the DRT plot

1. Click the DRT tab.
2. The current GUI provides several DRT display choices: Gamma vs Tau, Gamma vs Frequency, g vs Tau, and g vs Frequency.
3. For a conventional first inspection, Gamma vs Tau is a useful default view.
4. Count the number of clearly visible DRT peaks that you want to deconvolute. Do not automatically choose a large number of peaks just to force a better-looking decomposition.

Step 8 - Perform peak deconvolution

1. In the Peak Analysis section, enter the Number of peaks visible in the DRT curve.
2. Under Peak deconvolution, click Run.
3. In some cases, if the deconvolution does not start correctly or the DRT settings were changed, run Simple Run once more and then run Peak deconvolution again.
4. After a successful deconvolution, the individual peak curves are displayed and a Peak Parameters table appears below the plot.

The peak table includes values such as peak ID, series name, peak area, resistance contribution, relaxation time (τ), capacitance, peak frequency, and FWHM. These values can then be exported for further analysis or reporting.

Interpretation caution: A mathematical peak is not automatically a unique electrochemical mechanism. Assigning a peak to charge transfer, gas diffusion, adsorption, transport, contact effects, or another process requires independent physical and experimental evidence.

Step 9 - Export the results

The Export Result panel provides three main output actions:

- DRT -> Export: saves a CSV file named “[original file name] DRT.csv”.
- EIS Regression -> Export: saves “[original file name] EIS regression.csv”.
- Figure -> Save: saves a 300-dpi PNG named “[original file name] figure.png”. If peak deconvolution has been completed, the program also saves “[original file name] parameters.csv” automatically.

The peak-parameter CSV includes the extracted peak information, including resistance, relaxation time, capacitance, peak frequency, and FWHM. Choose an output folder that you can easily find later, such as a dedicated Results folder next to the original EIS file.

Step 10 - A recommended routine for every dataset

1. Import one EIS dataset.
2. Confirm the active-area setting.
3. Verify the Nyquist plot.
4. Run Simple Run with the default settings.
5. Check the EIS regression fit.
6. Adjust regularization/FWHM only when needed, and rerun.
7. Inspect the DRT curve and decide how many peaks are physically defensible.
8. Run peak deconvolution.
9. Review the peak-parameter table.
10. Export DRT, EIS regression, figure, and peak parameters.
11. Record the settings used so the analysis can be reproduced later.

11. Contact and citation

Contact

For questions, comments, or technical issues regarding this software:

Masood Fakouri Hasanabadi

Email: fakourih@ualberta.ca

Repository: <https://github.com/MFakouri/pyDRT-Peak-Analysis->

How to cite this software

If you use pyDRT-Peak-Analysis in research, presentations, reports, or publications, cite the software repository and the relevant original pyDRTtools method papers.

1. Open the GitHub repository page: <https://github.com/MFakouri/pyDRT-Peak-Analysis->
2. On the right side of the repository page, find “Cite this repository”.
3. Open it and copy the citation format required by your journal or reference manager. GitHub generates this information from the repository CITATION.cff file.

Version consistency note: At the time this guide was prepared, setup.py reported software version 0.4.0, while CITATION.cff reported version 0.1.0. Before formally releasing or publishing the guide, the repository maintainer should align these version numbers so users receive a consistent citation.

Citation currently generated from CITATION.cff

Fakouri Hasanabadi, M. (2026). pyDRT-Peak-Analysis (Version 0.1.0) [Computer software]. GitHub.
<https://github.com/MFakouri/pyDRT-Peak-Analysis->

Suggested wording in a Methods section

DRT analysis and peak deconvolution were performed using pyDRT-Peak-Analysis (Fakouri Hasanabadi, 2026), a modified and extended version of pyDRTtools. The relevant original pyDRTtools/DRTtools methodology was also cited according to the analysis features used.

Original pyDRTtools/DRTtools references to cite when relevant

Feature used	Reference
Basic DRT analysis / RBF discretization	Wan, T. H., Saccoccio, M., Chen, C., & Ciucci, F. (2015). Influence of the discretization methods on the distribution of relaxation times deconvolution: implementing radial basis functions with DRTtools. <i>Electrochimica Acta</i> , 184, 483–499. https://doi.org/10.1016/j.electacta.2015.09.097
Regularization-parameter selection	Maradesa, A., Py, B., Wan, T. H., Effat, M. B., & Ciucci, F. (2023). Selecting the Regularization Parameter in the Distribution of Relaxation Times. <i>Journal of The Electrochemical Society</i> , 170, 030502. https://doi.org/10.1149/1945-7111/acbca4

Bayesian credible intervals	Ciucci, F., & Chen, C. (2015). Analysis of electrochemical impedance spectroscopy data using the distribution of relaxation times: A Bayesian and hierarchical Bayesian approach. <i>Electrochimica Acta</i> , 167, 439–454. https://doi.org/10.1016/j.electacta.2015.03.123
Bayesian/hierarchical Bayesian regularization	Effat, M. B., & Ciucci, F. (2017). Bayesian and hierarchical Bayesian based regularization for deconvolving the distribution of relaxation times from electrochemical impedance spectroscopy data. <i>Electrochimica Acta</i> , 247, 1117–1129. https://doi.org/10.1016/j.electacta.2017.07.050
Hilbert-transform / EIS quality analysis	Liu, J., Wan, T. H., & Ciucci, F. (2020). A Bayesian view on the Hilbert transform and the Kramers-Kronig transform of electrochemical impedance data: Probabilistic estimates and quality scores. <i>Electrochimica Acta</i> , 357, 136864.

Minimum citation recommendation: For a standard Simple Run + peak-deconvolution analysis, cite the pyDRT-Peak-Analysis software repository and the foundational DRTtools/pyDRTtools paper by Wan et al. (2015). Add the other references only when you use the corresponding methods.

End of guide